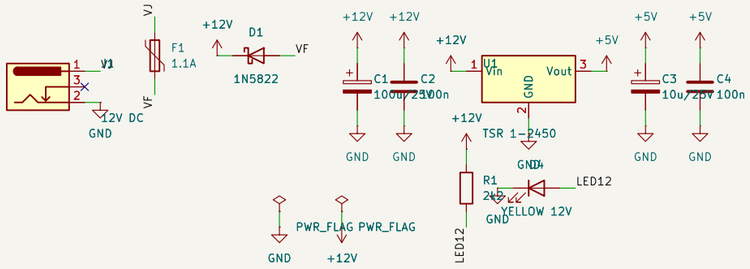
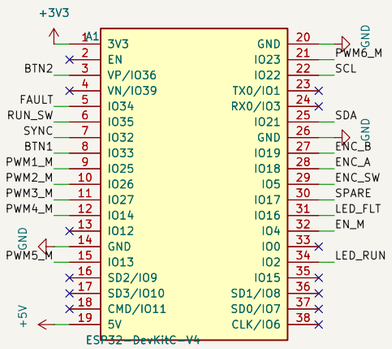
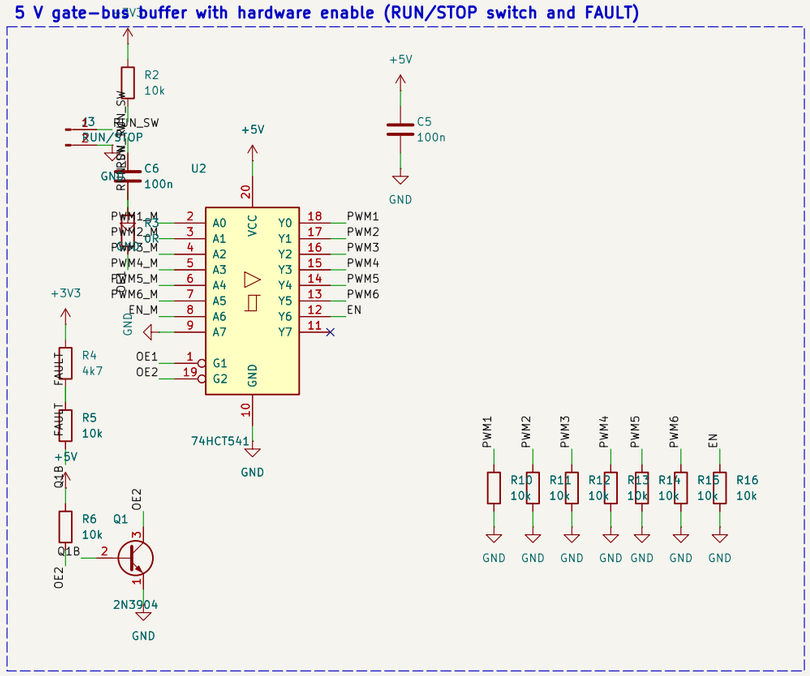
Power: 12 V in -> 5 V (TSR 1-2450); 12 V also feeds the gate bus



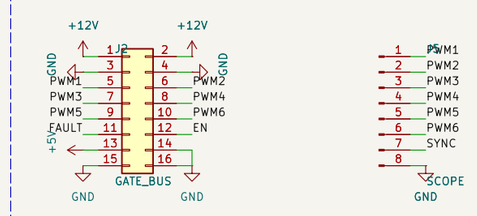
ESP32-DevKitC V4 (socketed, programmed over its own USB)



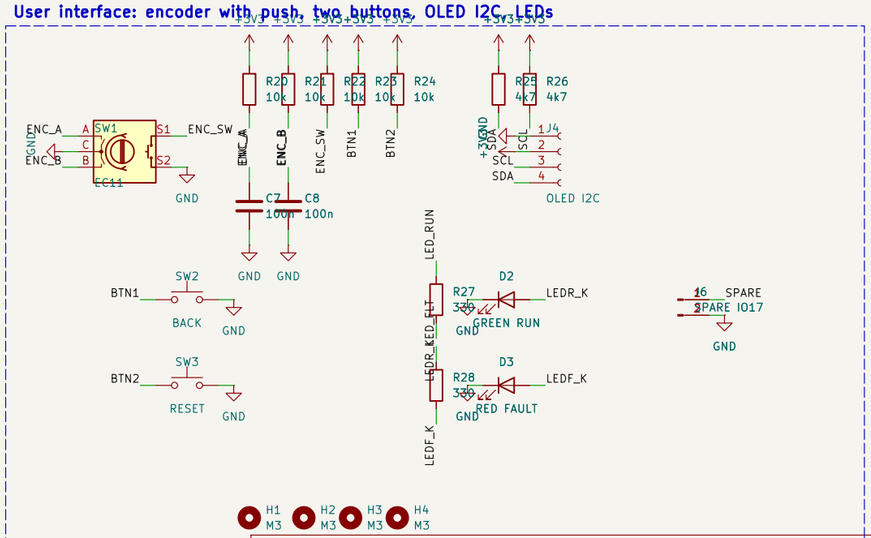
5 V gate-bus buffer with hardware enable (RUN/STOP switch and FAULT)



Gate bus (IDC-16) and scope header



User interface: encoder with push, two buttons, OLED I2C, LEDs



Gating block: ESP32 PWM generator with 5 V buffered gate bus

UPB — Power Electronics

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File: gating_block.kicad_sch

Title: Gating block (ESP32 DevKitC + 74HCT541 gate bus driver)

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 1/1